Master's Thesis: Status and Plan

Biofilm growth coupled into the IKM FEM element, following Klempt et al.

Keisuke Nishioka

Institute of Continuum Mechanics (IKM), Leibniz Universität Hannover
Keio University

19 October 2026

Since the last meeting: the scope of the FEM part is fixed

Thesis

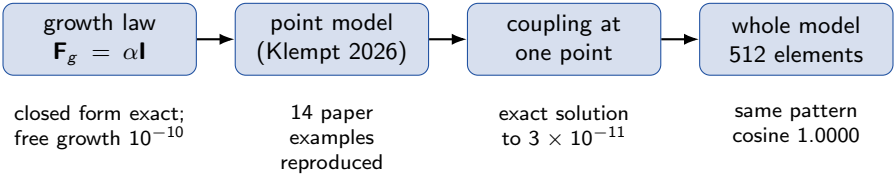
- **one and two species**
- coupled into the IKM element (Klempt et al. 2024)
- composition from the point model (Klempt et al. 2026)

Continuation at Keio

- more species (from four)
- two-way coupling

One and two species can be checked against published examples.

Verification, step by step



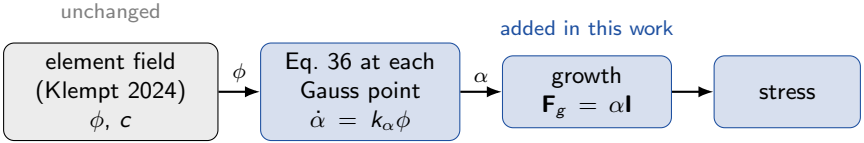
Each step is checked before the next one builds on it.

- point model: two misprints in Klempt et al. 2026 identified (no $+\gamma$ in Eq. 17, no $\frac{1}{2}$ in Eq. 19)
- same point model in Fritsch et al. 2025: their figures reproduced within 0.013

The coupling: what the element does and what is added

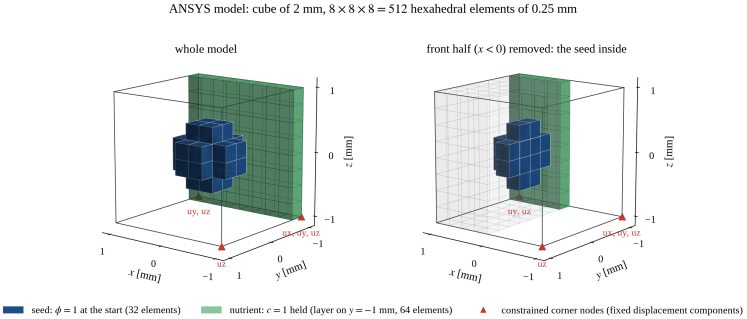
The element solves the biofilm field of Klempt et al. 2024 (Eq. 34):

$$\dot{\phi} - \beta\,\Delta\phi - k_\alpha\,\alpha + |\nabla\phi|\,r\,\frac{c}{k+c}\,\mathbf{n}_\phi \cdot \mathbf{n}_c = 0$$



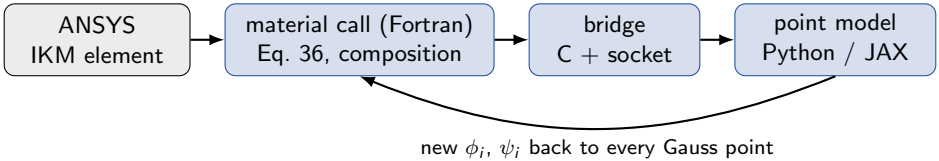
Klempt 2024 Table 2: $k_\alpha = 10^{-3}$, $E = 10$ Pa, $\nu = 0.49$. β, r, k, d, g: the framework's example input, not from the paper.

Model and boundary conditions



- 2 mm cube, 512 elements; only rigid-body motion fixed
- nutrient $c = 1$ on the face $y = -1$ mm; seed of 32 elements with $\phi = 1$
- uniform growth is stress free; stress comes from differences in growth
- used for Checks 1–3 and the two-species runs (ANSYS); the Klempt 2024 reproduction uses the paper's 20 μm cube (Python)

What I implemented



speed single-element run 357 s → **5.6 s**

check every ANSYS call replayed in Python, difference 0

used as it is

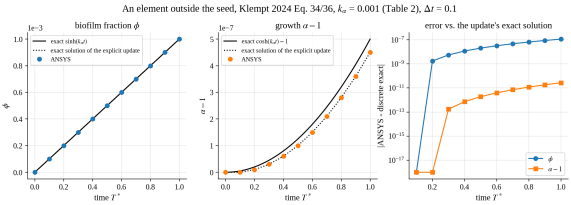
- IKM element: field, stiffness
- point-model equations (Klempt et al. 2026)

written in this work

- Eq. 36 and the composition step in the material call
- Fortran → Python bridge, sub-stepping
- point-model solver (JAX): all 14 paper examples
- checks: closed form in ANSYS, call replay, about 440 tests

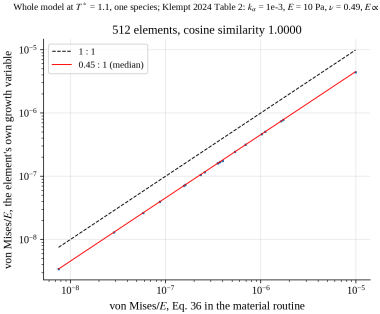
Checks 1 and 2: exact solution and whole model

1. Exact solution



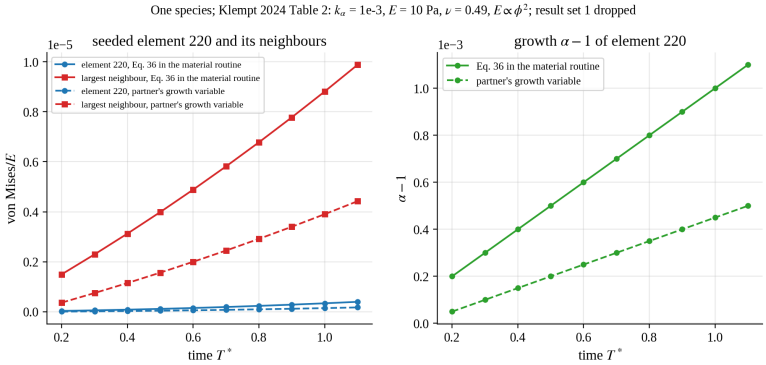
α matches to 3×10^{-11}

2. Whole model, 512 elements



same pattern (cosine 1.0000); factor $0.455 = \frac{1}{2} \cdot \frac{10}{11}$

Check 3: the seeded element and its neighbours



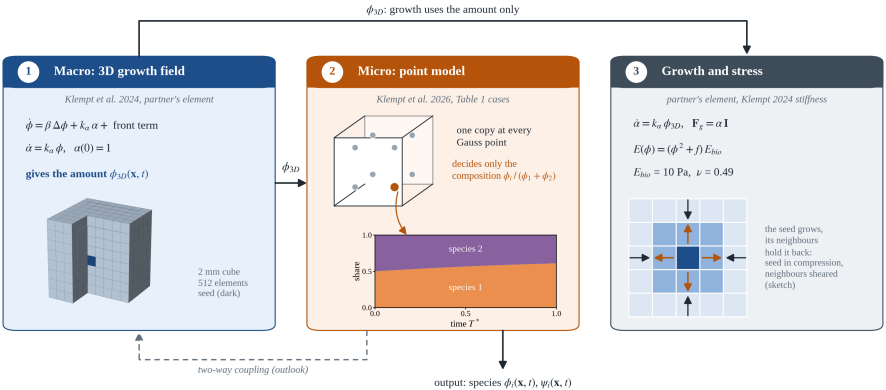
- $\alpha - 1 = 1.1 \times 10^{-3} = k_\alpha \phi t$, as Eq. 36 says
- seed compressed; neighbours carry about 25× its von Mises stress
- von Mises/$E \leq 10^{-5}$: slow growth (stresses scale with E)

Four things found along the way (partner element)

1. growth variable = mean of both species fields $\Rightarrow$ halved with one species
   the unused field stays at its start value; intended?
2. start values must satisfy $\phi_1 + \phi_2 \leq 1$
   the example input starts both at 1; for one species I start the second at 0
3. first result set not in equilibrium $\Rightarrow$ excluded
   zero stress on the first calls while the element sets up its operators
4. front-growth term inactive (reading slip) $\Rightarrow$ the seed does not spread
   it also uses $|\Delta\phi|$ where Eq. 34 has $|\nabla\phi|$; corrected in a test copy, ϕ exceeds 1

Questions for Oliver. Checks 1–3 do not depend on them.

Two species: composition from the point model (Klempt et al. 2026)



Point model run two ways: A) rescaled every step to ϕ_{3D}; B) started when ϕ_{3D} first reaches ϕ_{min}, then independent. Assumed, not from a paper: $s = 0.15$, $\phi_{cap} = 0.9$, $\phi_{min} = 0.01$.

- the field sets the **amount**, the point model the **composition**
- one-way: the composition does not act on the stress

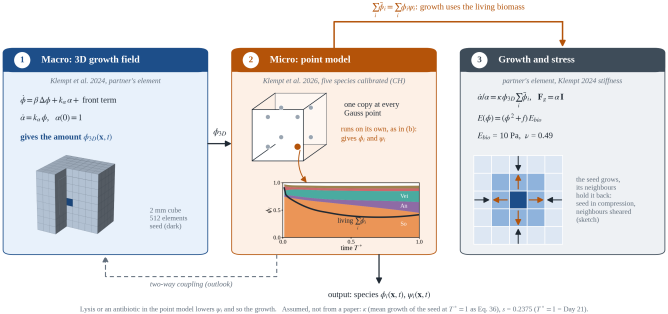
Two species: results in the packed seed (ANSYS, $T^* = 1$)

case (Klempt et al. 2026)	paper	coupled, $\phi_{\mathrm{cap}} = 0.9$	$\phi_{\mathrm{cap}} = 0.85$ / 0.95
3: coexistence	0.60	0.667	0.668 / 0.664
6: one species wins	≈ 0 ($\psi_1 = 0.05$)	0.021 ($\psi_1 = 0.048$)	–

share $\phi_1/(\phi_1 + \phi_2)$; paper: point model alone; parameters $c^* = 100$, $\alpha^* = 10$

- same type of outcome; the ratio shifts because the field sets the amount
- converged in the coupling time step

Added since 5 October: growth from the living biomass



Lysis or an antibiotic in the point model lowers ψ_i and so the growth. Assumed, not from a paper: κ (mean growth of the seed at $T^* = 1$ as Eq. 36), $s = 0.2375$ ($T^* = 1$ = Day 21).

Eq. 36 uses the amount: $\dot{\alpha} = k_\alpha \phi$.
Here: $\dot{\alpha}/\alpha = \kappa \, \phi_{3D} \sum_i \phi_i \psi_i$, so lysis in the point model lowers the growth.

κ chosen so that the mean growth of the seed equals Eq. 36 (an assumption, not from a paper; Appendix E).

seed, $T^* = 1$	Eq. 36	living	diff.
$\alpha - 1$, 8^3	$6.752 \cdot 10^{-4}$	$6.752 \cdot 10^{-4}$	0.0 %
von Mises/E, 8^3	$4.684 \cdot 10^{-5}$	$4.357 \cdot 10^{-5}$	−7.0 %
von Mises/E, 16^3; 24^3			−7.1; −7.8 %
antibiotic vs. none, 8^3	0 %	−32 %	

five species calibrated (CH), $E = 10$ Pa; 24^3 and the antibiotic case with the previous calibration

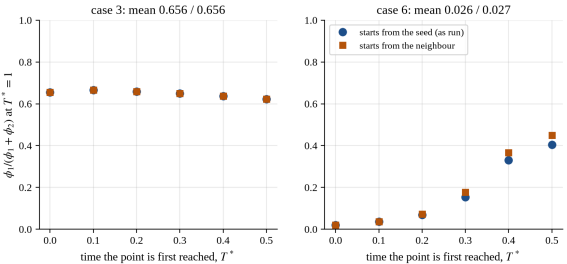
Two modelling assumptions

1. **time link** $s = 0.15$: point model advances $s\,\Delta t$ per step (no paper links the two time scales)
   case 6: winner appears for $s \gtrsim 0.15$; case 3: share moves by 0.03–0.06
2. **cap** $\phi_{\mathrm{cap}} = 0.9$ on the amount seen by the point model
   case 3: 0.664–0.668 for $\phi_{\mathrm{cap}} = 0.85$–$0.95$

All other parameters are the published values.

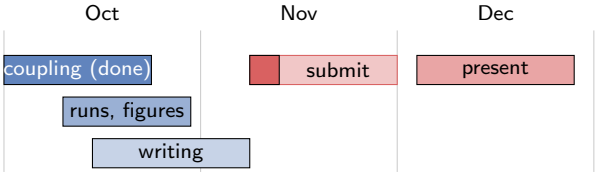
Limits

- composition is local, not carried with the front: effect on the mean ≤ 0.001 (right, Python)
- s has no source in a paper
- Eq. 34 as printed moves the colony instead of growing it (Appendix D)
- two α: the element's own and Eq. 36 are integrated separately (Check 2)



Composition at $T^* = 1$ of the points first reached at each time; Klempt 2024 test case 4.1 field (Python, 20 μm cube), Klempt 2026 cases 3 and 6.

Status and timeline



- admission granted 18 Aug 2026; examiners Prof. Junker (IKM), Dr. Wangenheim (IDS); supervisors Prof. Soleimani, Dr. Geisler; thesis in English
- NIFE internship completed (04.05–24.07), report submitted
- **submission in November** (aiming for early November); presentation in December, date to be fixed with both examiners

Questions for this meeting

1. **Scope:** one and two species in the thesis, more at Keio?
2. **Assumptions:** s and ϕ_{cap} with sensitivity studies acceptable?
3. **Keio:** anything needed for the double degree this term?
4. **Klempt et al. 2024:** form of Eq. 34/35, t_{ref}, unit of μ, Eq. 36 or Table 1?
5. **Structure:** TMCMC calibration, a short internship section, then the ANSYS coupling?
6. **Submission:** printed copies, electronic version, Keio side?

References

- Klempt, Soleimani, Junker (2023), PAMM: precursor spatial model
- Klempt, Soleimani, Wriggers, Junker (2024), Biomech. Model. Mechanobiol.: growth field, Eq. 34/36, Table 2
- Klempt, Geisler, Soleimani, Junker (2026), Arch. Appl. Mech.: multi-species point model, Table 1 cases
- Fritsch et al. (2025), arXiv:2512.15145: Bayesian updating, same model

Appendix A: notation

Growth field and mechanics (Klempt et al. 2024)

ϕ	biofilm volume fraction (0 = void, 1 = dense)
c	nutrient concentration
α	expansion variable, $\mathbf{F}_g = \alpha \mathbf{I}$, $\alpha(0) = 1$; growth is $\alpha - 1$
$\mathbf{F} = \mathbf{F}_e \mathbf{F}_g$	elastic / growth split
k_α	growth factor (10^{-3} per T^*)
β	phase-field regularisation
r, k	front growth rate, Monod constant
T^*	normalised time, $T^* \in [0, 1]$

Point model (Klempt et al. 2026)

ϕ_i	volume fraction of species i
ϕ_0	void, $\phi_0 = 1 - \sum_i \phi_i$
ψ_i	living fraction (viability) of species i
γ	Lagrange multiplier for $\sum \phi_i + \phi_0 = 1$
A_{ij}	interaction matrix (self, cooperation, competition)
b_i	sensitivity to the antibiotic
c^*, α^*	nutrient, antibiotic concentration
η_i	viscosity (time scale) of species i

Coupling (this work)

ϕ_{3D}	amount from the element field
$\phi_i / \sum_j \phi_j$	composition (share of species i)
s	time link: point model advances $s\,\Delta t$
ϕ_{cap}	largest amount the point model sees

Appendix B: background, growth field and mechanics

Kinematics. Growth is a volume change: $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$ with isotropic $\mathbf{F}_g = \alpha\mathbf{I}$. Stress comes only from $\mathbf{F}_e$ (neo-Hookean). A body that can grow freely stays stress free; stress appears where growth is constrained.

Material (Klempt et al. 2024, Eq. 20 and Table 2): $E = 10$ Pa, $\nu = 0.49$, stiffness weighted with ϕ^2 (void: floor $10^{-3}E$); no viscosity. Stresses scale linearly with E (the example input's 1000 MPa gives $10^8\times$ larger values).

Field equations (Klempt et al. 2024, from an extended Hamilton principle):

$$\dot{\phi} = \underbrace{\beta\,\Delta\phi}_{\text{diffusion}} + \underbrace{k_\alpha\,\alpha}_{\text{local source}} - \underbrace{r\,\frac{c}{k+c}\,\nabla\phi\cdot\frac{\nabla c}{|\nabla c|}}_{\text{front growth towards nutrient}}\,, \qquad \dot{\alpha} = k_\alpha\,\phi$$

- ϕ and c are nodal fields of the element; α is an internal variable at the Gauss points
- The front term moves the biofilm boundary up the nutrient gradient; it vanishes where ϕ or c is uniform
- The nutrient diffuses much faster than the biofilm grows (quasi-static c)

Appendix C: point model and one substep of the coupled run

Point model (Klempt et al. 2026), per species i, no space dependence, implicit, $\Delta t = 10^{-4}$:

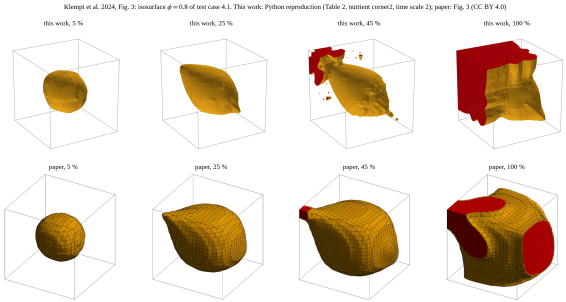
$$(\eta_{\phi,i} + \eta_i \psi_i^2)\,\dot{\phi}_i + \eta_i \phi_i \psi_i\,\dot{\psi}_i = c^* \psi_i \textstyle\sum_j A_{ij}\phi_j\psi_j - \gamma - (\text{barrier})$$

$$\eta_i\,(\phi_i \psi_i\,\dot{\phi}_i + \phi_i^2\,\dot{\psi}_i) = c^* \phi_i \textstyle\sum_j A_{ij}\phi_j\psi_j - \alpha^* b_i\,\psi_i - (\text{barrier}), \qquad \textstyle\sum_i \phi_i + \phi_0 = 1$$

One substep $t_n \to t_{n+1}$: ANSYS solves only $\operatorname{div}\boldsymbol{\sigma} = \mathbf{0}$; the material routine turns $\mathbf{F}$ into $\boldsymbol{\sigma}$ after removing the growth.

repeat (Newton iteration of ANSYS)
  $\mathbf{F}$ at every Gauss point from the current displacements
  for each Gauss point
    $\phi_{3D} = \phi_1 + \phi_2$ from the element; restore the converged state of t_n
    $\alpha_{n+1} = \alpha_n + k_\alpha\,\phi_{3D}\,\Delta t$ Eq. 36
    $\mathbf{F}_e = \mathbf{F}\,\mathbf{F}_g^{-1}$, $\mathbf{F}_g = \alpha_{n+1}\mathbf{I}$ $\to$ $\boldsymbol{\sigma}$, tangent
    if two species and $\phi_{3D} \geq \phi_{\min}$:
      scale the point model to $\min(\phi_{3D}, \phi_{\mathrm{cap}})$
      bridge (Fortran $\to$ C $\to$ socket $\to$ Python): n_{sub} steps of $s\Delta t/n_{\mathrm{sub}}$
      scale back; store ϕ_i, ψ_i, γ as state variables
    return $\boldsymbol{\sigma}$ and the tangent to ANSYS
  assemble the residual, update the displacements
until converged
the element advances ϕ_1, ϕ_2, c to t_{n+1} (its own solver)

Appendix D: Klempt et al. 2024 test cases, reproduced in Python



test case	time scale	RMS ϕ	RMS c
4.1 (Fig. 4)	2	0.03	0.02
4.2 “high” (Fig. 7)	10	0.04	0.04
4.2 “low” (Fig. 7)	4–5	0.09	0.05

Independent solver on the paper's 20 μm cube, not the ANSYS model. Table 2 unchanged; growth on both faces and consumption $g\phi c$ (two departures from Eq. 34/35 as printed); nutrient in a 2 μm corner block; one time scale per run ($T^* = t/t_{\mathrm{ref}}$, Sec. 4). Curves agree; the shape agrees in kind (egg towards the corner) but is more slender, also with other diffusion and a 2nd-order scheme. Table 3's pressure fits μ in MPa.

Appendix E: growth from the living biomass, implementation and runs

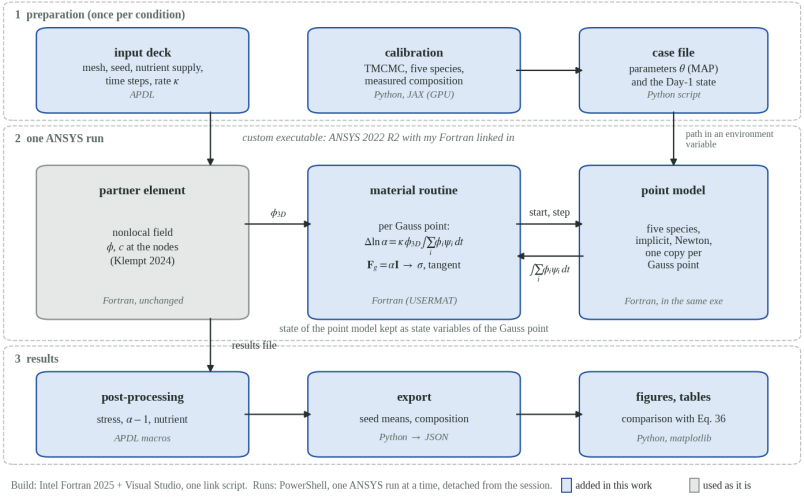
$$\ln \alpha_{n+1} = \ln \alpha_n + \kappa\, \phi_{3D} \int_{t_n}^{t_{n+1}} \sum_i \phi_i \psi_i \, dt \qquad \text{(Appendix C, line 5 only)}$$

item	how
form	increment of $\ln \alpha$, as Klempt 2024 Table 1
$\phi_{3D} \in [0, 1]$	only marks where the biofilm is
point model	one per Gauss point, on its own (not scaled to ϕ_{3D})
start	when $\phi_{3D} \geq \phi_{\min}$, from the calibrated Day-1 state
integral	over its own substeps, clock s per unit T^*
storage	state variables of the Gauss point; element unchanged
κ	$8.14 \cdot 10^{-3}$: mean $\alpha - 1$ as Eq. 36 ($-0.003\,\%$); assumed

seed, $T^* = 1$	Eq. 36	living	diff.
$\alpha - 1$, 8^3	$6.752 \cdot 10^{-4}$	$6.752 \cdot 10^{-4}$	$0.0\,\%$
von Mises/E, 8^3	$4.684 \cdot 10^{-5}$	$4.357 \cdot 10^{-5}$	$-7.0\,\%$
vM, 16^3; 24^3			-7.1; $-7.8\,\%$
vM/($E(\alpha - 1)$)	0.069	0.064	
antibiotic, 8^3	$0\,\%$	$-32\,\%$	

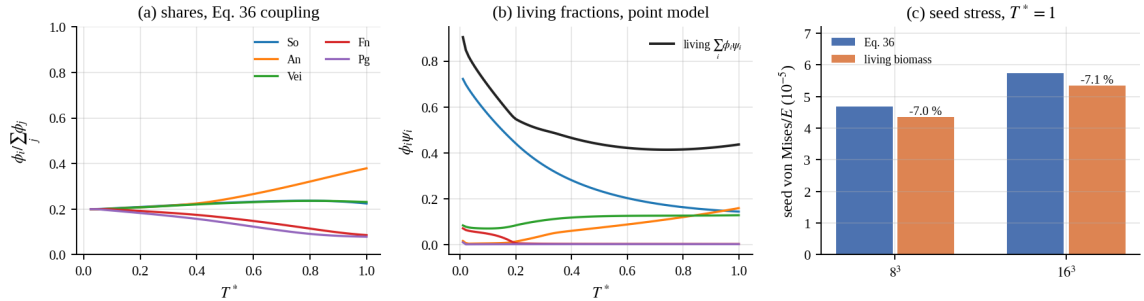
CH, $E = 10$ Pa, $\nu = 0.49$; 24^3 and the antibiotic case with the previous calibration. Same amount of growth, different spread inside the seed; $\alpha - 1$ and stress scale with κ.

Appendix F: how the run is implemented, and the tools



Blue: added in this work; grey: used as it is. One line of the material routine changes compared with Eq. 36.

Appendix G: calibrated five-species model in the element



	start	$T^* = 1$	growth $\alpha - 1$	seed von Mises
shares (Eq. 36)	0.2 each	An 0.38, Vei 0.23, So 0.23, Fn 0.09, Pg 0.08	$6.752 \cdot 10^{-4}$, same as two species	vM$/E = 4.68 \cdot 10^{-5}$
living biomass $\sum_i \phi_i \psi_i$	0.90	0.44 (mostly So lost)	same mean	$-7.0\,\%$ (8^3), $-7.1\,\%$ (16^3)

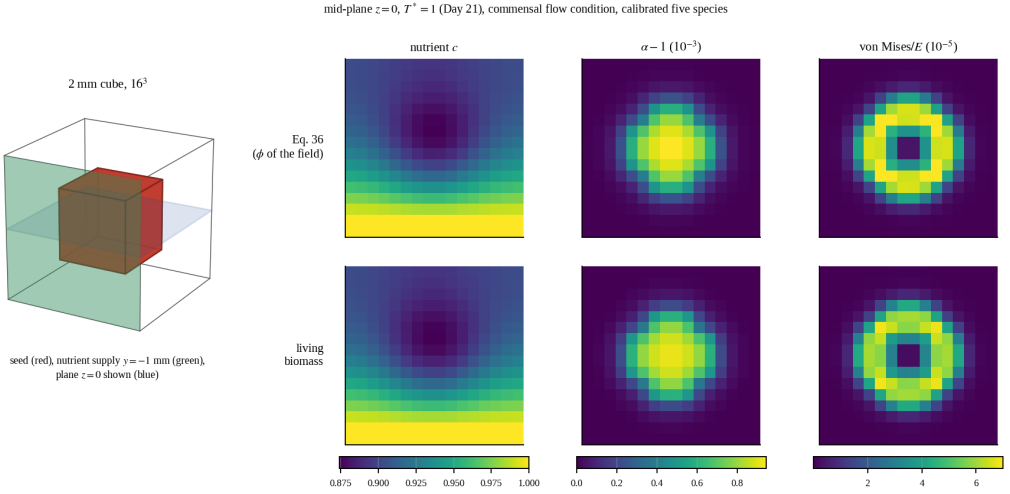
commensal flow condition (CH), calibration to the measured data, $E = 10$ Pa, 8^3

Appendix H: two and five species, implementation

	two species (5 October)	five species (since 10 October)
point model	Python, through a material server (Fortran $\rightarrow$ C $\rightarrow$ socket $\rightarrow$ Python)	Fortran, in the same executable (no server, about 4 times faster, same results)
parameters	Klempt et al. 2026 cases 3 and 6 ($c^* = 100$, $\alpha^* = 10$)	calibration (TMCMC, MAP, 5×5 interactions), $c^* = 25$, $\alpha^* = 0$
start	equal shares	equal shares; calibrated Day-1 state for the living-biomass growth
stiffness	partner's example input, $E = 1000$ MPa	Klempt 2024, $E = 10$ Pa
coupling (Eq. 36)	same code: amount scaled to ϕ_{3D}, $\phi_{\mathrm{cap}} = 0.9$, time link $s = 0.15$	
growth from the living biomass	–	yes; point model runs on its own, $s = 0.2375$ ($T^* = 1$ is Day 21)

The number of species is a parameter (1 to 5) of the same routine.

Appendix I: fields on the mid-plane, five species



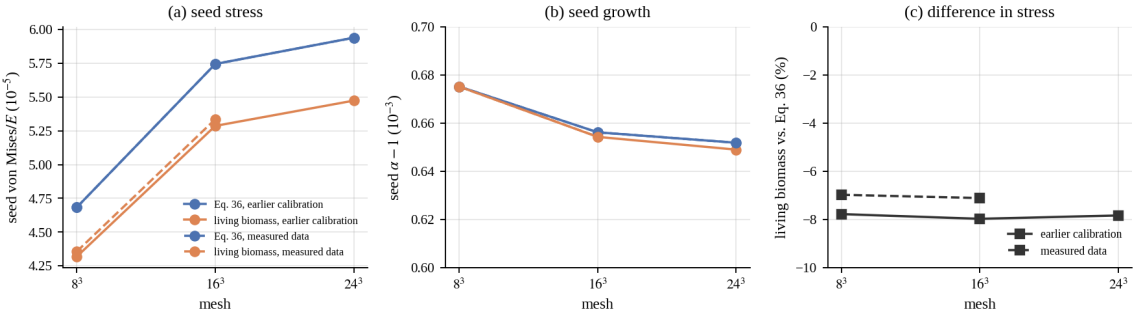
Element values at $T^* = 1$, 16^3. The two growth laws give the same nutrient and almost the same $\alpha - 1$; the stress ring around the seed is lower with growth from the living biomass.

Appendix J: analysis conditions

geometry, mesh	2 mm cube; 8^3 (512), 16^3, 24^3 hexahedral elements; seed of 32 elements at the centre (8^3, refined with the mesh)
mechanical BC	only rigid-body motion fixed (corner nodes); otherwise traction free
nutrient BC	$c = 1$ on the face $y = -1$ mm; no flux elsewhere
initial state	$\phi = 1$ in the seed, ≈ 0 outside; $\alpha = 1$; $c = 1$ everywhere
time	$T^* = 1$ is Day 21; 95 steps ($\Delta T^* = 1/95$); ANSYS 2022 R2, static, nonlinear geometry
material	neo-Hooke, $E = 10$ Pa, $\nu = 0.49$, $E \propto \phi^2$ (Klempt 2024 Table 2); $\mathbf{F}_g = \alpha \mathbf{I}$
growth field	$k_\alpha = 10^{-3}$, $\beta = 0.02$ mm$^2/T^*$ (Table 2 scaled to the 2 mm cube, assumed)
point model	two species: Klempt 2026 cases 3 and 6; five species: calibrated MAP (CH), $c^* = 25$
coupling	$\phi_{\mathrm{cap}} = 0.9$, $\phi_{\mathrm{min}} = 0.01$, time link $s = 0.15$ (assumed); living-biomass growth: $s = 0.2375$, $\kappa = 8.14 \cdot 10^{-3}$ (assumed)
output	seed means of $\alpha - 1$ and von Mises$/E$ at $T^* = 1$

Values not from a paper are marked as assumed.

Appendix K: mesh convergence



- seed stress: $+23\,\%$ from 8^3 to 16^3, $+3.4\,\%$ from 16^3 to 24^3; converging
- seed growth changes by less than $3.5\,\%$
- the difference between the two growth laws is -7 to $-8\,\%$ on every mesh, so 8^3 is enough to compare them
- measured-data calibration at 24^3 with growth from the living biomass: still running